

RAG Evaluation Cheatsheet

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* | Measure, diagnose, and improve Retrieval-Augmented Generation (RAG) performance with *TruLens*.

The 3 Core Metrics

Metric	Checks	Nickname
Groundedness	Are all answers based on retrieved data?	<i>Hallucination Detector</i>
Context Relevance	Did we retrieve the <i>right</i> data?	<i>Retriever Health Check</i>
• Answer Relevance	Did we answer the <i>actual</i> question?	<i>Response Quality Check</i>

1. Groundedness (Faithfulness) — “The Hallucination Detector”

* **Definition:**
Measures whether the generated answer is *factually supported* by the retrieved context — no hallucinations or made-up facts.

Example

Context Retrieved:
> “The Eiffel Tower was completed in 1889 and stands 330 meters tall.”

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Good (Grounded):

> “The Eiffel Tower is 330 meters tall.”

Bad (Not Grounded):

> “The Eiffel Tower is 330 meters tall and was designed by Gustave Eiffel in collaboration with Leonardo da Vinci.”

Why Bad:

Hallucinates information not supported by the context.

2. Context Relevance (Retrieval Quality) — “Right Data, Right Answer”

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Definition:

Measures whether the retrieved documents or chunks actually contain *information relevant* to answering the user’s question.

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Example

User Question:

> “What are the health benefits of green tea?”

Good (Relevant Context):

- “Green tea contains antioxidants called catechins that may reduce inflammation.”
- “Studies show green tea consumption is associated with lower cardiovascular disease risk.”

Bad (Irrelevant Context):

- “The tea ceremony in Japan dates back to the 9th century.”
- “Green tea is grown primarily in China, Japan, and India.”

Why Bad:

These contexts don’t help answer the question about health benefits.

3. Answer Relevance (Response Quality) — “On Topic or Not?”

Definition:

Measures whether the generated answer actually *addresses the user’s question* clearly and directly.

Example

User Question:

> "How do I reset my password?"

Good (Relevant Answer):

> "To reset your password, click 'Forgot Password' on the login page, enter your email, and follow the link sent to your inbox."

Bad (Irrelevant Answer):

> "Passwords are important for security. Strong passwords should contain uppercase, lowercase, numbers, and symbols."

Why Bad:

Talks about passwords generally but doesn't answer *how to reset* it.

How the Three Metrics Work Together

graph TD

A[User Query] --> B[Retrieval: Context Relevance]

B --> C[Generation: Groundedness]

C --> D[Final Answer: Answer Relevance]

Each stage builds on the previous: - **Context Relevance** → Did we get the *right data*?

- **Groundedness** → Is the answer *supported by that data*?

- **Answer Relevance** → Does the response *actually answer* the question?

End-to-End Example

Query:

> "What is the refund policy for online orders?"

1. Retrieval:

→ "Online orders can be returned within 30 days for a full refund."

High Context Relevance

2. Generation:

→ "You can get a full refund if you return within 30 days."

High Groundedness

3. **Final Answer:**

→ Directly answers the refund question.
High Answer Relevance

Result:

All three metrics high → **High-quality RAG response!**

**Fix Order: Context → Groundedness
→ Answer**

Bad retrieval breaks everything downstream.

Good RAG = Right Data (Context Relevance) + No
Hallucinations (Groundedness) + On Topic (Answer
Relevance)

Scoring Quick View

Metric	Ship It	Tune It	Fix It	Stop
Groundedness	0.90-1.0	0.70-0.89	0.50-0.69	<0.50
Context Relevance	0.80-1.0	0.60-0.79	0.40-0.59	<0.40
Answer Relevance	0.90-1.0	0.70-0.89	0.50-0.69	<0.50

Note: Thresholds should be set *case-by-case* based on your application’s goals and tolerance for error.

Quick Wins

If Groundedness is low:

→ Add to prompt: *“Base your answer ONLY on the provided context. If unsure, say ‘I don’t have enough information.’”*

If Context Relevance is low:

→ Improve retrieval configuration, embedding quality, or metadata filtering to return more relevant context.

If Answer Relevance is low:
→ Add to prompt: *“Answer the question directly and concisely. Stay focused on what was asked.”*

Rules of Thumb

- 1. **You can’t improve what you don’t measure.**
 - 2. **Fix retrieval first** — garbage in = garbage out.
 - 3. **All three must be high** for reliable RAG evaluation.
 - *4. **Monitor continuously** — data and model drift can affect scores.
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Common Score Patterns

Pattern	Meaning	Fix
All high	Ready for prod	Deploy & monitor
All low	System broken	Debug fundamentals
Low context	Retrieval issue	Tune embeddings or search
Low groundedness	Hallucination	Add stricter prompts
Low answer	Prompting	Simplify and focus

Readiness Checklist

- ☐ Metrics meet internal thresholds
 - ☐ Tested on diverse queries
 - ☐ Monitoring and alerts in place
 - ☐ Escalation path defined
 - ☐ Baseline metrics tracked
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Evaluate your RAG system with confidence — measure retrieval quality, factual grounding, and response accuracy using TruLens.